CIS-445 Machine Learning
Weka – Individual Project 3 Report

Student Name
Dylan Shea

The tutorial is worth 100 points.

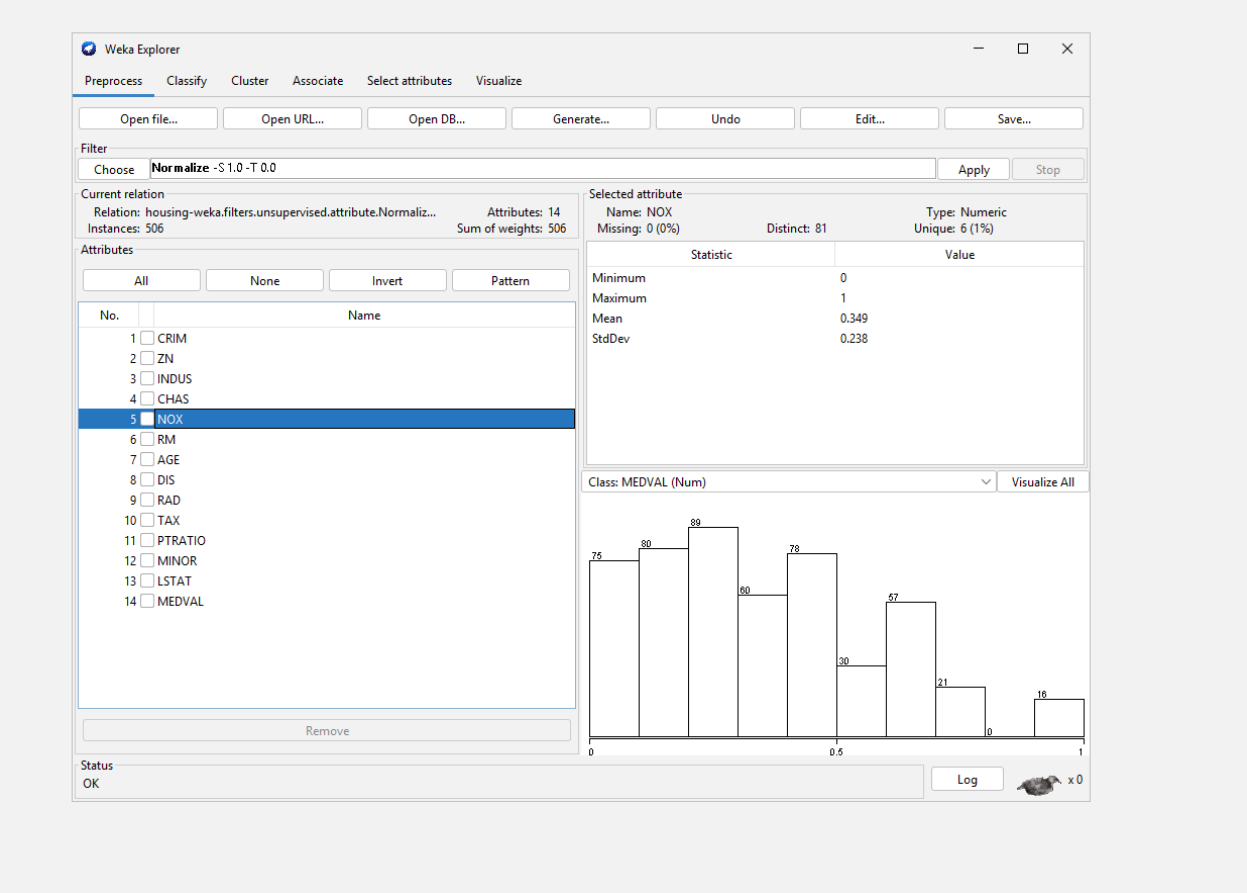
Objectives: Learn about Weka, the Machine Learning Workbench

This project contains two parts: Part A and Part B. In both parts you will use Weka Explorer and Weka Experimenter. In Part A you will work with the `housing.arff` dataset and essentially follow the given instructions. In Part B, however, you will work with `Salary.arff` dataset and explore the performance of several models on your own. Some parts of the project describe Weka and machine learning algorithms. You need to read these parts. There are also other parts which you have to work hands-on, step by step, in Weka. Note that you need to copy some of the results and answer the questions which I ask in the project and paste them into this document so that I can check how well you followed all steps in the project. Because this is the second version of the Weka – Individua Project 3, I would appreciate your comments and suggestions regarding the organization, clarity, spelling, content, and/or anything else.

Please note that this file which now contains 3 pages may grow to a few more pages after you copy and paste here the screen images, tables, answers to questions, etc. from Weka – Individual Project 3 file.

Page 7

Copy and paste Figure 6 below. The Figure should be <u>from your run</u>, not from my run.

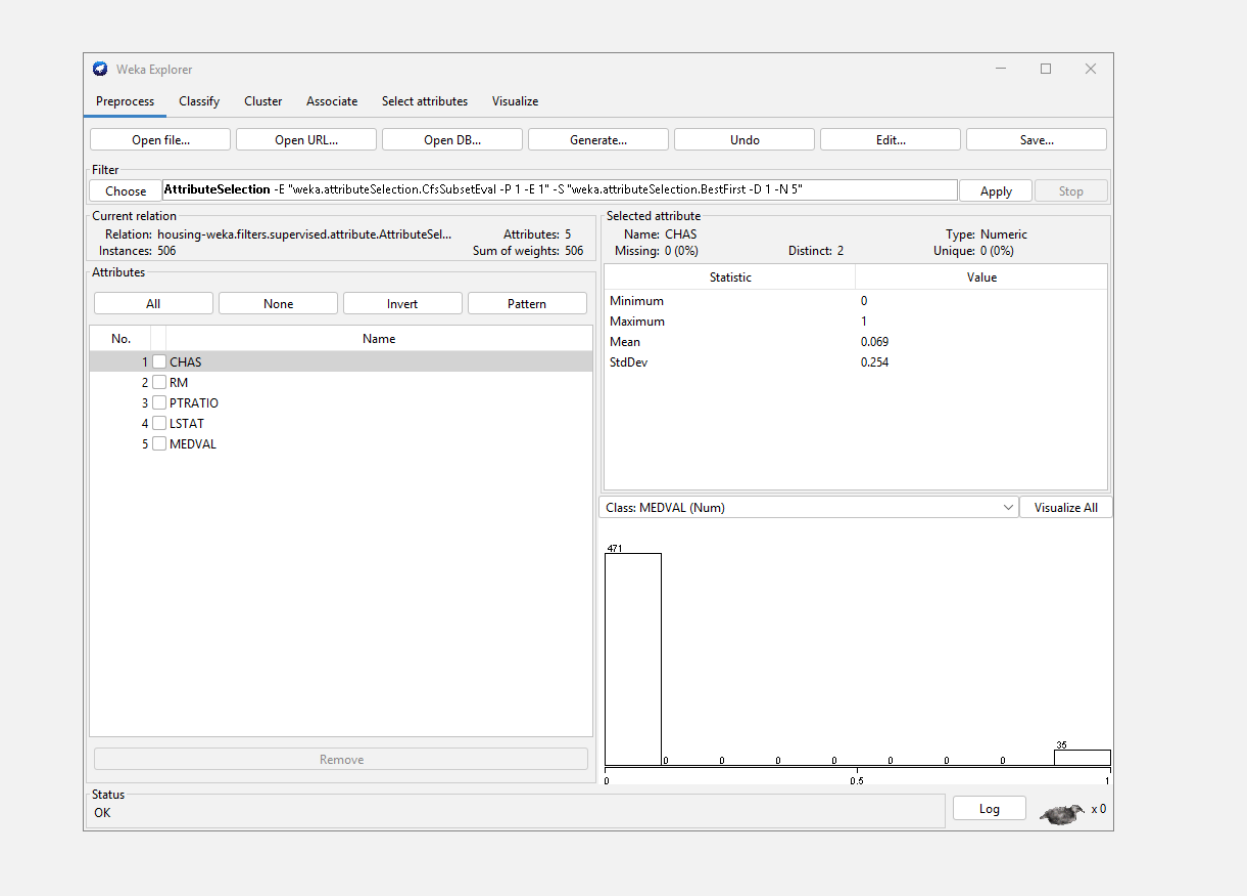


Do all numeric attributes are mapped into the [0,1] range? Yes

How about the MEDVAL attribute? Is it normalized too? No

Page 10

Copy and paste Figure 8 here. The Figure should be <u>from your run</u>, not from my run.

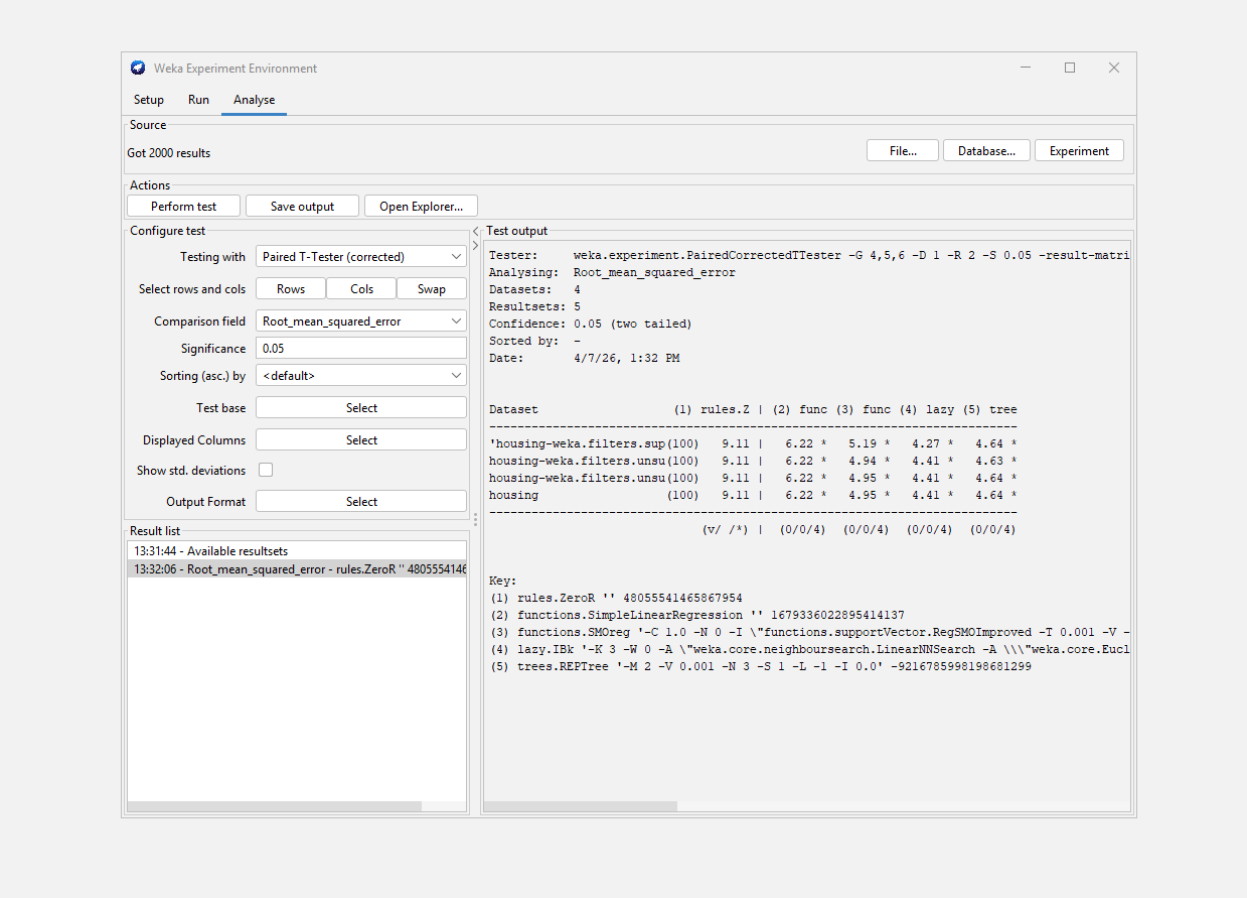


How many and which attributes were retained by the attribute selection filter? 5, Chas, RM, PTRatio, LSTAT, MEDVAL

List the retained attributes below.

Page 13

Copy and paste Listing 1 here. The listing should be <u>from your run</u>, not from my run.



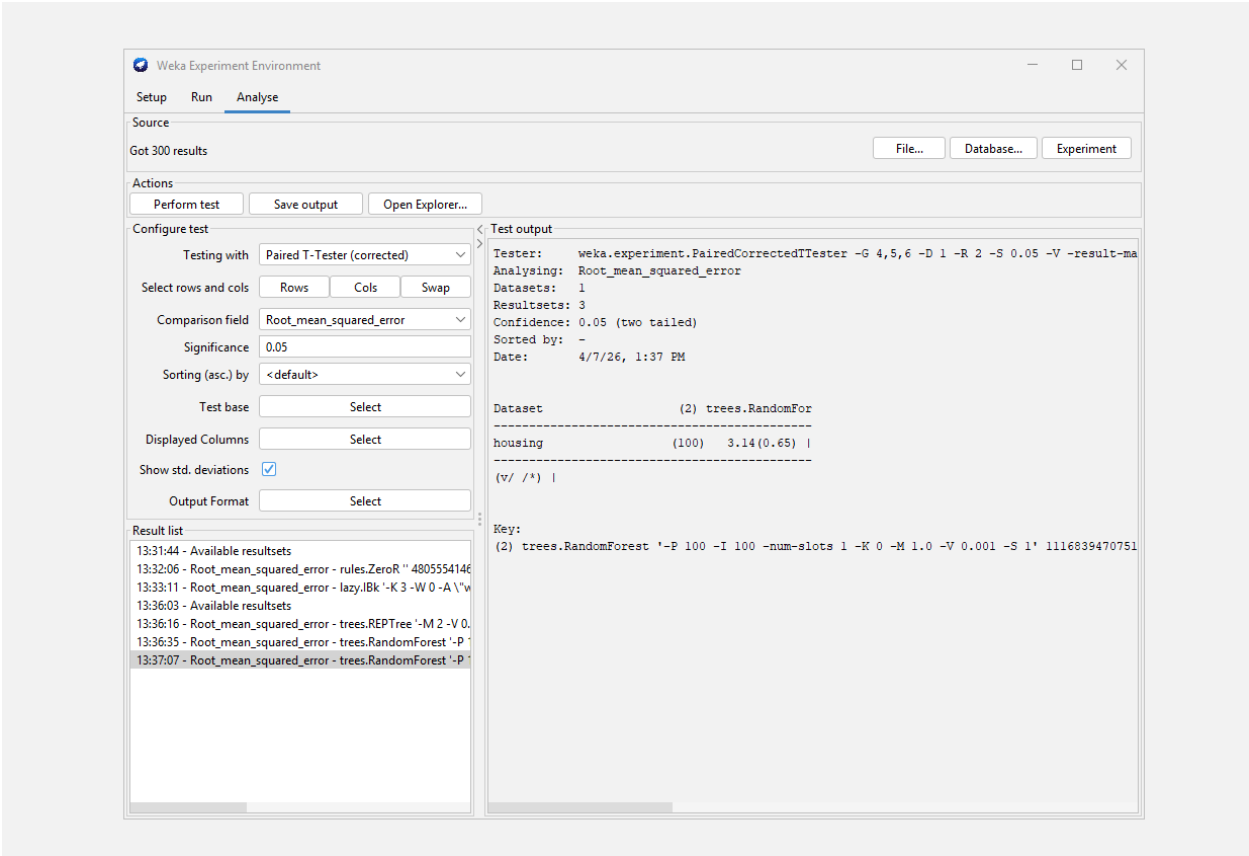
In Listing 1 which algorithm is used as the baseline to which the remaining algorithms are compared? ZeroR

How did the other 4 algorithms do in terms of performance (RMSE) compared to the baseline algorithm? Did they do statistically better, worse or the same? Remember the objective in regression is to have the RMSE as small as possible. Comment.

They perform statstically better than ZeroR.

Page 25

Copy and paste Listing 9 here. The listing should be <u>from your run</u>, not from my run.



Which model turned out to be the best on terms of RMSE? Random Foest

What is the value of RMSE and its standard deviation? 3.14, 0.65

Page 28

Discuss briefly the distribution of attributes.
Relatively left leaning for most of the attributes, higher salaries are more rare.

Did you normalize numeric attributes or standardize them?

I standardized

Which attributes were retained by the attribute selection filter? List them below.
Gender, Education Level, Job, Years of Expierence, Race, Salary

Did you do the test runs of Linear Regression, IB*k*, Bagging, M5P Tree, and Random Forest in Weka Explorer?
I did not,

What are the preliminary results in terms of the performance with respect to the correlation coefficient (R), root mean squared error (RMSE) and mean absolute error (MAE)? Discuss.

Across the 3 datasets, it was clear that the linear regression performed the worst, while Both of the datasets in the tree set performed the best, with the highest correlation and lowest errors.

Compare the performance of IB*k*, Bagging, M5P Tree, and Random Forest to *Linear Regression* (benchmark) with respect to the Correlation Coefficient (CC), RMSE, and MAE. Are the differences in the models' performance statistically significant at p=0.05?

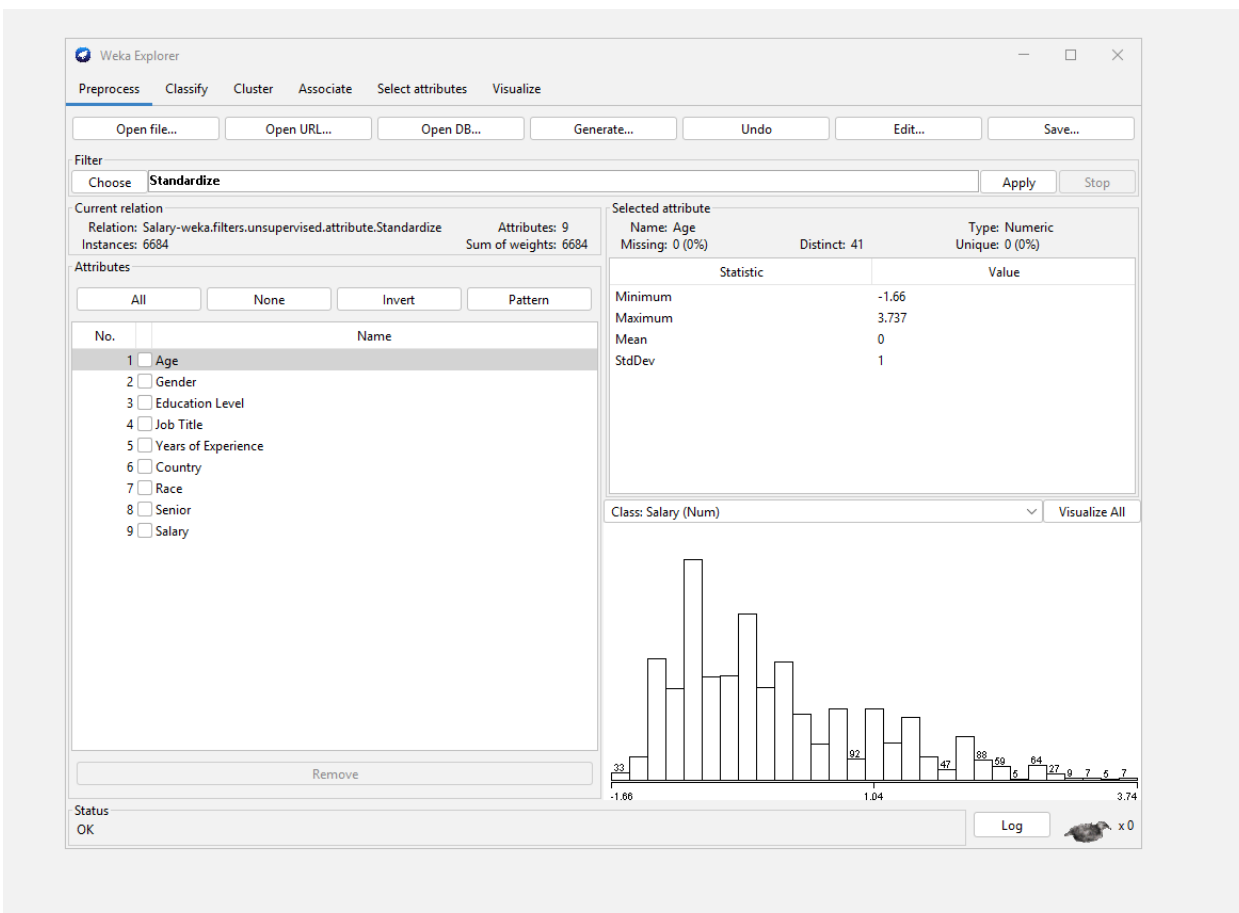
All of the models are statstically significantly better than Linear Regression other than IBK.

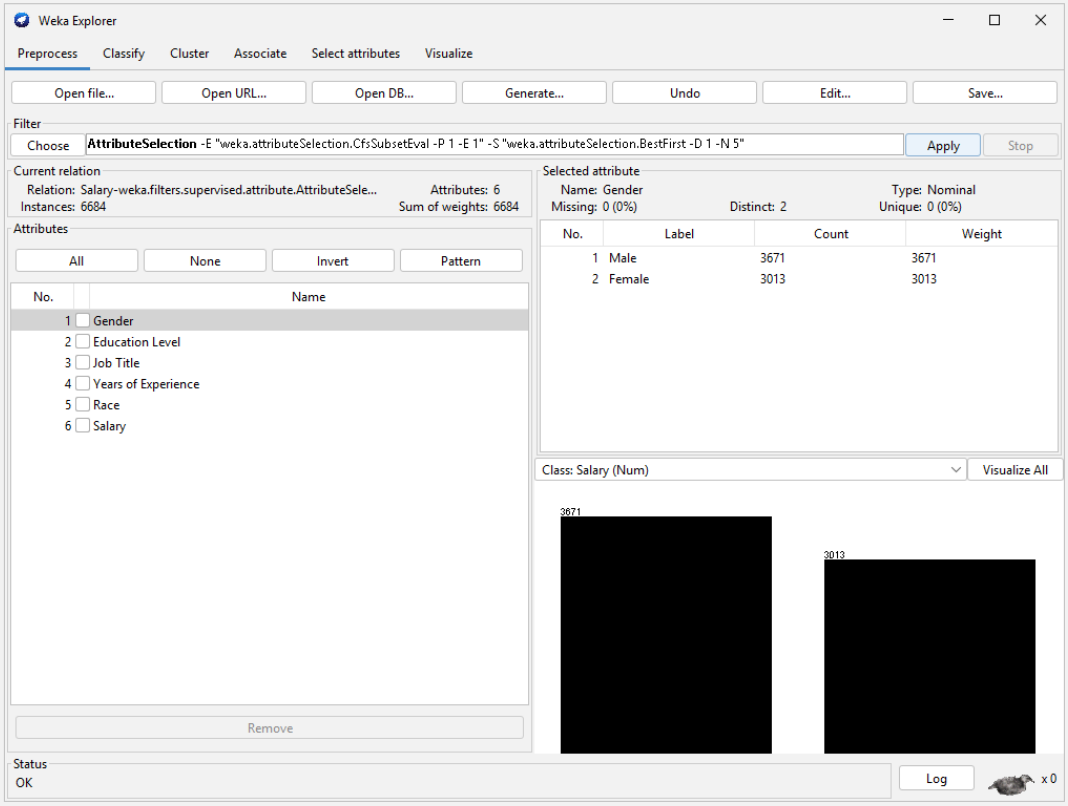
Do normalization or standardization, or attribute selection which you applied, improve the performance of the models?

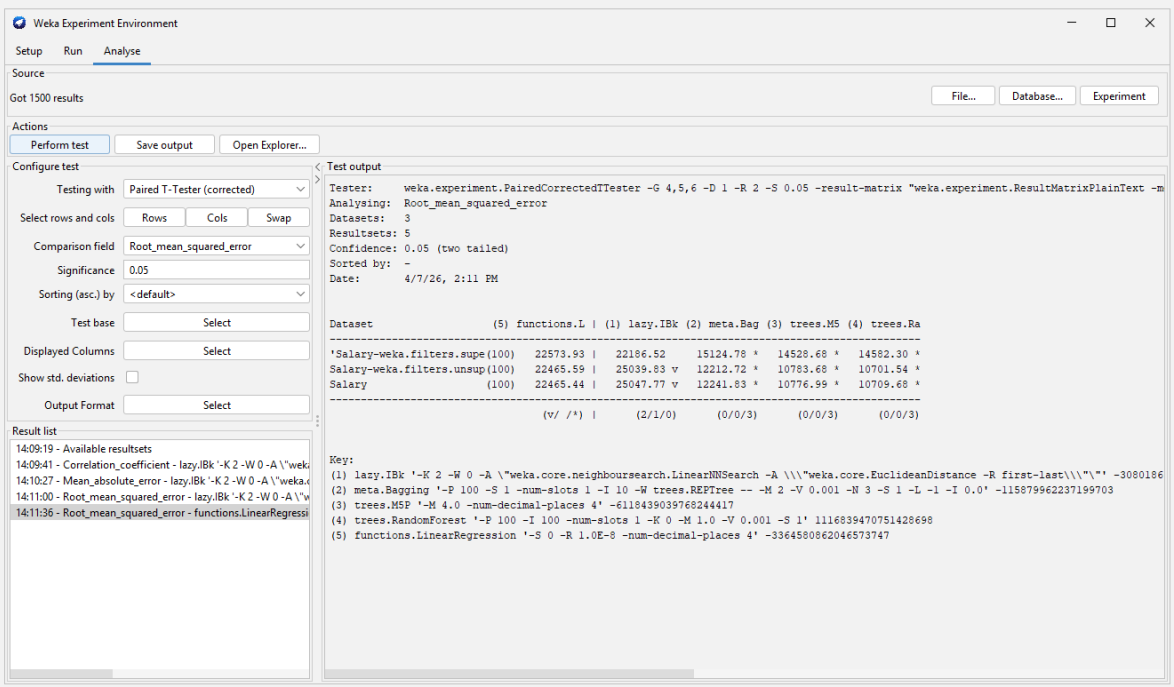
Yes, it does, it increase stability and accuracy by reducing outleirs.

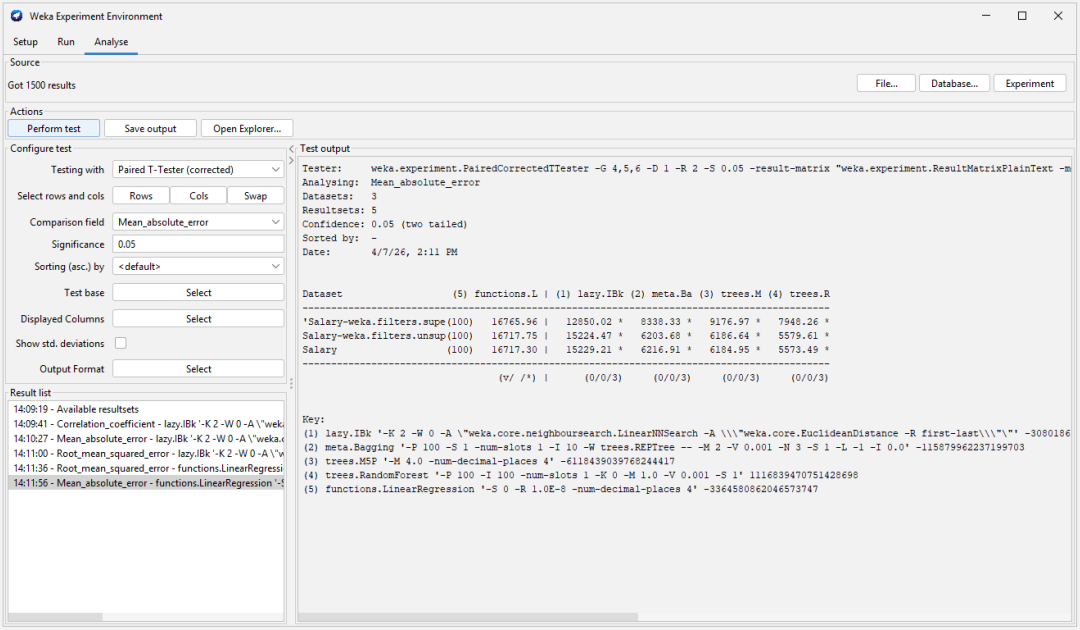
Discuss Part B of the project and the results.
I created the approipriate datasets(Reg, Standardize, Attribute, Selection)
I then saved them, and ran the expierement, which took over 20 minutes.
I reran the expiement with all 3 of the required stats, and then took screenshots, and answered the questions.

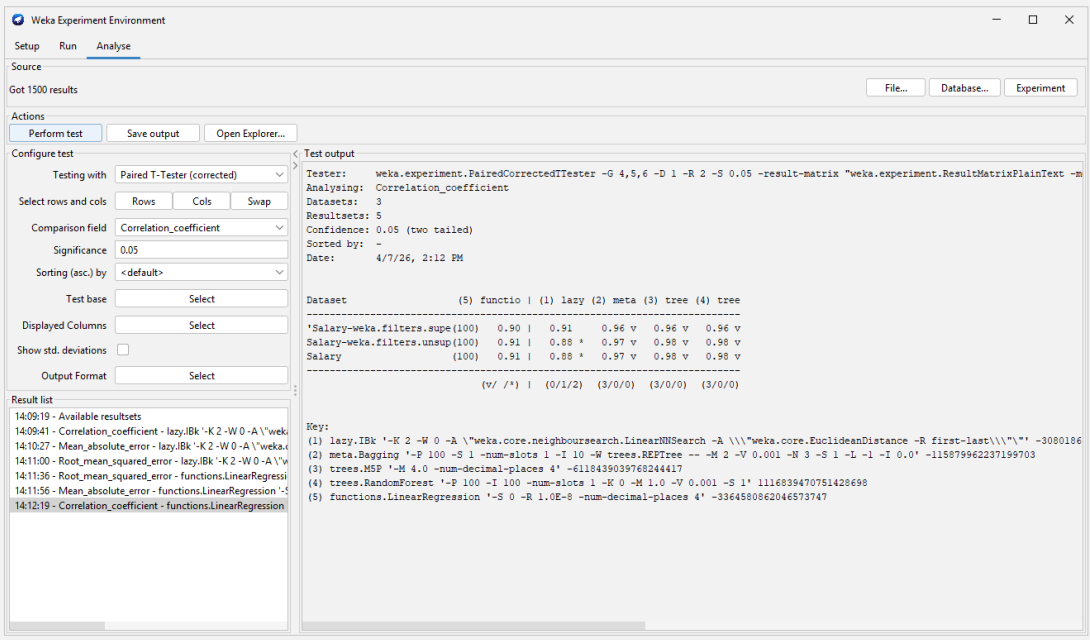
Copy and paste all relevant screen shots from Weka Experimenter below.




Weka Explorer
Preprocess
Classify
Cluster
Associate
Select attributes
Visualize
Open file...
Open URL...
Open DB...
Generate...
Undo
Edit...
Save...
Filter
Choose
AttributeSelection -E "weka.attributeSelection.CfsSubsetEval -P 1 -E 1" -S "weka.attributeSelection.BestFirst -D 1 -N 5"
Apply
Stop
Current relation
Relation: Salary-weka.filters.supervised.attribute.AttributeSele...
Attributes: 6
Instances: 6684
Sum of weights: 6684
Selected attribute
Name: Gender
Type: Nominal
Missing: 0 (0%)
Distinct: 2
Unique: 0 (0%)
No. Label Count Weight
1 Male 3671 3671
2 Female 3013 3013
Attributes
All
None
Invert
Pattern
No. Name
1 Gender
2 Education Level
3 Job Title
4 Years of Experience
5 Race
6 Salary
Remove
Class: Salary (Num)
Visualize All
3671
3013
Status
OK
Log
x 0


Weka Experiment Environment
Setup
Run
Analyse
Source
Got 1500 results
File...
Database...
Experiment
Actions
Perform test
Save output
Open Explorer...
Configure test
Testing with
Paired T-Tester (corrected)
Select rows and cols
Rows
Cols
Swap
Comparison field
Root_mean_squared_error
Significance
0.05
Sorting (asc.) by
<default>
Test base
Select
Displayed Columns
Select
Show std. deviations
Output Format
Select
Result list
14:09:19 - Available resultsets
14:09:41 - Correlation_coefficient - lazy.IBk '-K 2 -W 0 -A \"weka
14:10:27 - Mean_absolute_error - lazy.IBk '-K 2 -W 0 -A \"weka.
14:11:00 - Root_mean_squared_error - lazy.IBk '-K 2 -W 0 -A \"w
14:11:36 - Root_mean_squared_error - functions.LinearRegressi
Test output
Tester: weka.experiment.PairedCorrectedTTester -G 4,5,6 -D 1 -R 2 -S 0.05 -result-matrix "weka.experiment.ResultMatrixPlainText -m
Analysing: Root_mean_squared_error
Datasets: 3
Resultsets: 5
Confidence: 0.05 (two tailed)
Sorted by: -
Date: 4/7/26, 2:11 PM
Dataset (5) functions.L | (1) lazy.IBk (2) meta.Bag (3) trees.M5 (4) trees.Ra
'Salary-weka.filters.supe (100) 22573.93 | 22186.52 15124.78 * 14528.68 * 14582.30 *
Salary-weka.filters.unsup (100) 22465.59 | 25039.83 v 12212.72 * 10783.68 * 10701.54 *
Salary (100) 22465.44 | 25047.77 v 12241.83 * 10776.99 * 10709.68 *
(v/ /*) | (2/1/0) (0/0/3) (0/0/3) (0/0/3)
Key:
(1) lazy.IBk '-K 2 -W 0 -A \"weka.core.neighboursearch.LinearNNSearch -A \\\"weka.core.EuclideanDistance -R first-last\\\"\"" -3080186
(2) meta.Bagging '-P 100 -S 1 -num-slots 1 -I 10 -W trees.REPTree -- -M 2 -V 0.001 -N 3 -S 1 -L -1 -I 0.0' -115879962237199703
(3) trees.M5P '-M 4.0 -num-decimal-places 4' -6118439039768244417
(4) trees.RandomForest '-P 100 -I 100 -num-slots 1 -K 0 -M 1.0 -V 0.001 -S 1' 1116839470751428698
(5) functions.LinearRegression '-S 0 -R 1.0E-8 -num-decimal-places 4' -3364580862046573747





Write your comments and suggestions below. (optional)